

[Eccles Building, U.S. Federal Reserve
Board of Governors](#)



ECON 100B: Macroeconomics

Class 23: Government
(Jones Chapter 18)

Ryan D. Edwards
April 21, 2026

Agenda

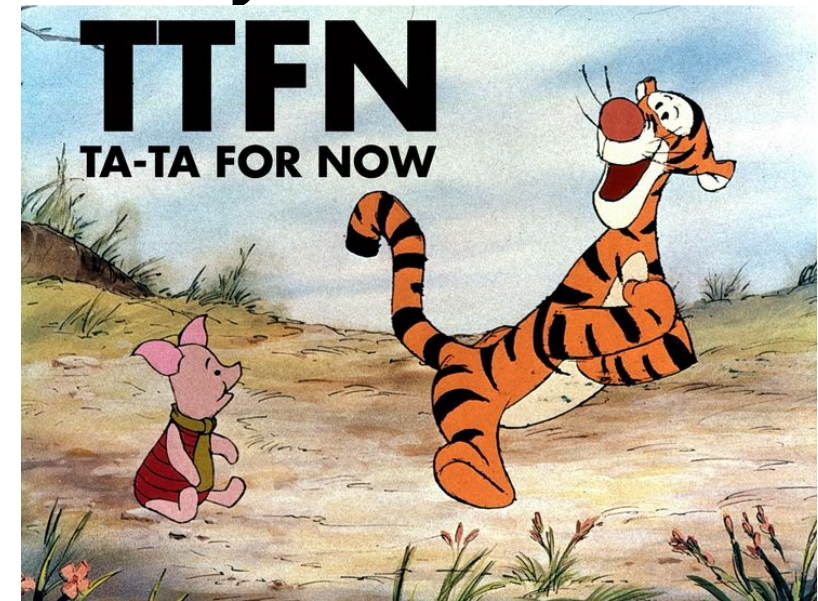
- Announcements, logistics
- Current events
- Chapter 18: Government

Same announcements

- Problem Set 3 is available, due on Friday, April 17
- **Problem Set 4** is available and is due on **Friday, May 1**
- All students get to drop their lowest PS score. If you have scored 100% on PS1, PS2, and PS3, you can skip PS4. But I don't recommend it
- The **Final exam** is confirmed for **Monday, May 11**
 - Just like at midterm, the main class meets here in **Pimentel 1**
 - We will run 90 minutes from about 3:00-4:30PM
 - DSP students: Please either make a proctoring appointment, or for 150% time only, we will have **Anthropology and Art Practice 160**
(I think it's kind of near Caffe Strada on Bancroft)

Current events: Ta-ta for now, tariffs

- The most wonderful thing about tariffs
Is that tariffs are wonderful things?
- On Monday 4/20/26, U.S. Customs and Border Protection reportedly began taking applications from importers for IEEPA tariff refunds following the [SCOTUS ruling](#)
- [NYT 4/20/26](#), note that consumers cannot apply for refunds
- With so much in the news, my money is on importers' filing for and pocketing the refunds quietly. But we shall see



Current events

- Fed Chair designee Kevin Warsh appeared before the Senate Banking Committee at 7am Pacific Time this morning
- [In prepared remarks,](#)
 - Warsh spoke highly of Chair Bernanke and the recovery from the Great Recession [when Warsh had been a Fed governor]
 - Spoke of his time at Stanford ... *It's not his fault that it wasn't Berkeley*
 - “[M]onetary policy independence is essential. Monetary policymakers must act in the nation’s interest . . . their decisions the product of analytic rigor, meaningful deliberation, and unclouded decision-making
 - “I do not believe the operational independence of monetary policy is particularly threatened when elected officials—presidents, senators, or members of the House—state their views on interest rates.
 - “Low inflation is the Fed’s plot armor, its vital protection against slings and arrows. So, when inflation surges—as it has done in recent years—grievous harm is done to our citizens, especially to the least well-off. They lose purchasing power. Their standard of living falls. They may also lose faith in our system of economic governance, raising doubts whether monetary policy independence is all it’s cracked up to be.”

In the days and weeks to come

- Recall Jay Powell's [January 11, 2026 statement](#) and the [January 12, 2026 statement by all living former Fed chairs](#) about the Justice Department's January 9 subpoena
 - “The reported criminal inquiry into Federal Reserve Chair Jay Powell is an unprecedented attempt to use prosecutorial attacks to undermine [the Fed’s] independence.
 - “This is how monetary policy is made in emerging markets with weak institutions, with highly negative consequences for inflation and the functioning of their economies more broadly.
 - “It has no place in the United States whose greatest strength is the rule of law, which is at the foundation of our economic success.”
- Last week, despite a judge’s quashing of the subpoena, [the President and the U.S. Attorney for DC indicated they will not stop](#)
- [Thom Tillis \(R-NC\) has vowed to deadlock any nominations in the Senate Banking Committee](#), which has [13 Republications and 11 Democrats](#)

Learning objectives for Chapter 18

- Good governments can live forever, in contrast with good humans, who live and then must eventually die and settle up
- Governments usually do not have to pay off creditors in full, but they must stay on a **fiscally sustainable path**
- Fiscal sustainability is a finite **debt-to-GDP ratio**, B/Y
- What B/Y ratio? Good question. We're not sure, but we can recognize unsustainable debt-to-GDP ratios when we see them
- A big problem is that governments facing government budget **deficits** might **print more money** to pay them, raising **inflation**
- Large countries like the U.S. can run budget deficits and finance them with foreign saving ... (*suspenseful music plays*)

Secondary learning objectives for Chapter 18

- Government spend on core activities in G — defense, police, fire, roads, K-12
And they also support elderly and needy populations with transfer payments
- Government deficits — spending more, taxing less — can shift the IS curve rightward and create booms, jobs, and inflation
- In many big countries and groupings, smaller units of government often face borrowing constraints
 - In the U.S., the 50 states typically have balanced budget amendments
 - In the Euro area, the [Stability and Growth Pact](#) requires countries to stick to a 3% deficit-to-GDP ratio and a 60% debt-to-GDP ratio
- Without these legal constraints, governments within a currency area might desire higher deficits, causing
 - A rightward shift of the IS curve and local jobs and happiness
 - Increased inflation $\Delta\pi > 0$ through the Phillips curve
 - Higher interest rate by the central bank's reaction to higher inflation π
 - At worst case, a bailout by the central bank of the member government might occur

Definitions

- It's important to differentiate between **stocks** and **flows**
- Like in the Solow model, where capital is a stock & saving is a flow
- **Taxes** or revenues = government's income
- **Spending** or expenditures = government's outflows or expenditures
- **Deficit** = the excess of spending over taxes
- **Surplus** = the excess of taxes over spending
- **Debt**, a stock = accumulated deficits
- **Assets**, a stock = accumulated surpluses

Government debt & deficits in the news in 2011

- On August 2, 2011, Congress and the President agreed to raise the **debt ceiling**, the legal limit on the total federal debt, \$14 trillion
- The deal set up *future* spending cuts of about \$1 trillion. These later became “the sequester” cuts that hit defense and discretionary spending in 2013
- Congress also formed a bipartisan “supercommittee” of charged with agreeing to plan to cut \$1.2 trillion more by November 23
- On August 5, 2011, Standard & Poor’s lowered the **credit rating** of the federal government anyway, from AAA to AA+ **for the first time ever**
- Over the weekend of November 19-20, it became clearer that the supercommittee would fail to agree on a plan
- On Monday, November 21, U.S. stock prices fell, but markets were also upset over European debt problems, which are worse

The size of government budgets in the U.S. depends on whether you mean total activity or the G in $Y = C + I + G + NX$

- We are primarily concerned with the U.S. federal government
 - Spending in 2025 around \$7.5 trillion in [NIPA Table 3.2](#), including big transfer payments like Social Security and Medicare
 - With GDP at \$30 trillion, broad federal government spending is 25% of GDP
- But state and local governments also tax and spend
 - California taxes income at a progressive marginal rate between 1% and 12.3% and [collected about \\$10K per person in 2022](#), or about 13% of [average personal income of \\$77K](#) because of other taxes
 - Across the nation, state and local budgets totaled about \$4.5 trillion in spending in 2025 in [NIPA Table 3.3](#)
- The G that goes into $Y = C + I + G + NX$ looks a little different, per [NIPA Table 1.1.5. In 2025,](#)

Federal $G = \$2$ trillion

State and local $G = \$3.3$ trillion

If state and local government budgets are just as big, why not examine them?

- Almost all states and localities have balanced budget amendments to their constitutions ([Rueben and Randall, Urban 2017](#))
- Some balanced budget amendments work better than others — California has traditionally been in fiscal trouble, with poor credit ratings on its debt
- California has majority-rule ballot propositions that can be anything, including extremely expensive things
- [Proposition 13 \(1978\)](#) also restricted property tax rates, which is a huge source of revenue for local governments
- But what about the federal government? It has no balanced budget amendment at all ...

Can the U.S. federal government just spend as much as it wants?

- All of us, including the government, have budgets
 - We spend money on housing, food, transportation, investment (like school) and hopefully some fun things too
 - We receive money from working, from family transfers, from asset income (bonds, stocks, trust funds)
 - We don't always have a balanced budget; we save when our incomes exceed our spending, and we also borrow, like when we go to college
- The government also spends, on things like defense and Social Security, and it receives money, called taxes
- It also saves or borrows — we call these surpluses and deficits
- Each of us knows we can't "live beyond our means" and borrow, or run deficits, forever. What about the government?

“Living within” or “beyond our means”

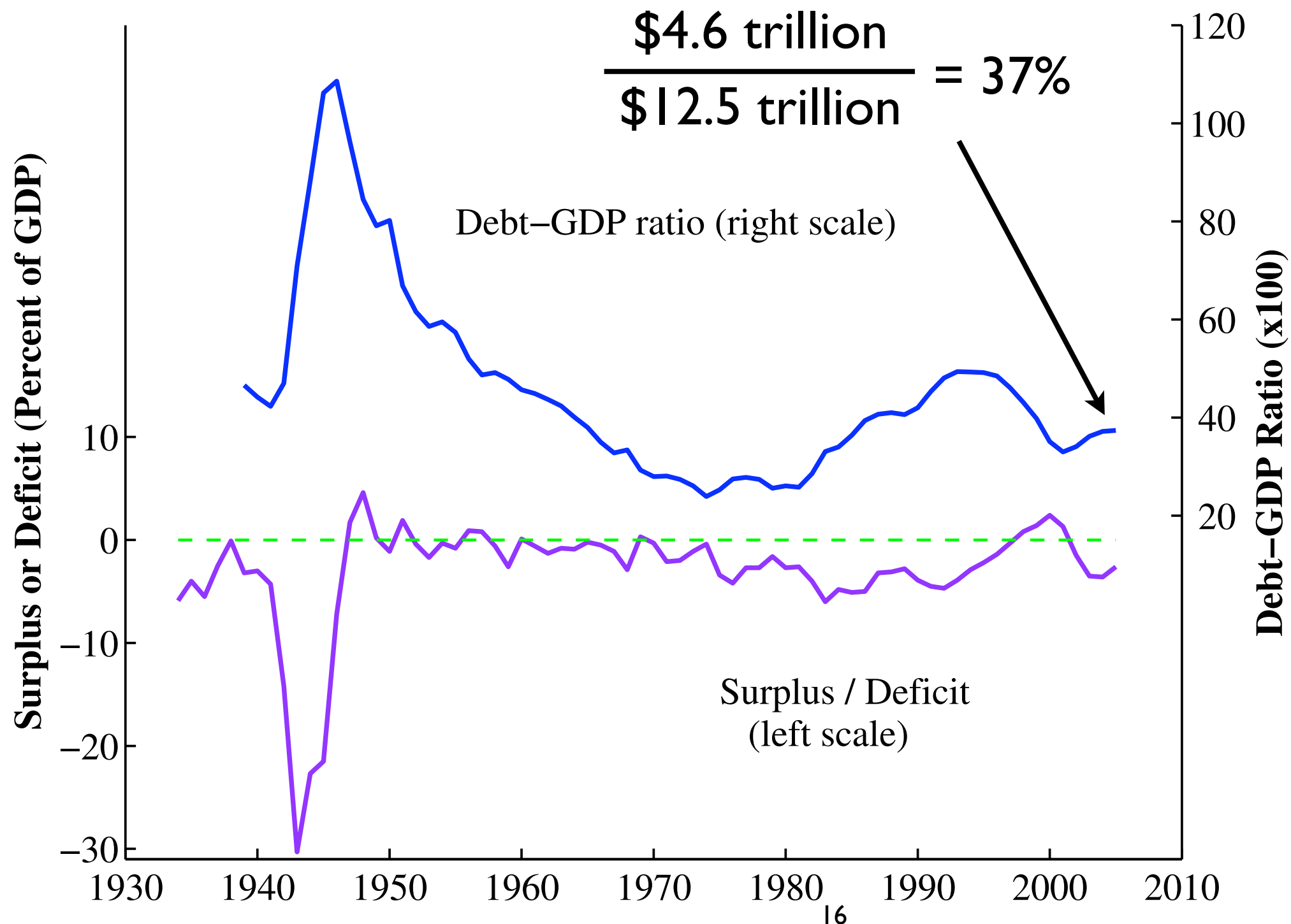
- When you fill out an application for a credit card, you include:
 - Your birthdate (which reveals your **age**)
 - Your annual **income**
 - Your Social Security Number
- Your credit limit depends on how much money you make (income), but also how much you are expected to make (age)
- What about your SSN? It wants to know your credit history: do you pay your debts? If not, no more loans!
- The government faces exactly the same kind of constraints: It must not borrow more than it can & will eventually pay off



What are the government's means? What determines whether it can pay its debts?

- **YOU DO.** You and I all pay taxes to the government, and the government can use that revenue to pay its debts
- But it doesn't stop there; **your children will too**, and your grandchildren will, and so on. Also, your parents and your grandparents
- As individuals, our means are more limited, although we receive **transfers**, gifts of money and time, from parents and grandparents
- But the federal government isn't like us — it will “live” forever
- *Changes of regime* don't change the government's commitment to paying its debts. Even after the Revolutionary War, the new U.S. government honored its war debts
- The U.S. government has never* missed a payment and has *stellar* credit

If the government is ageless, what statistic should determine its means?



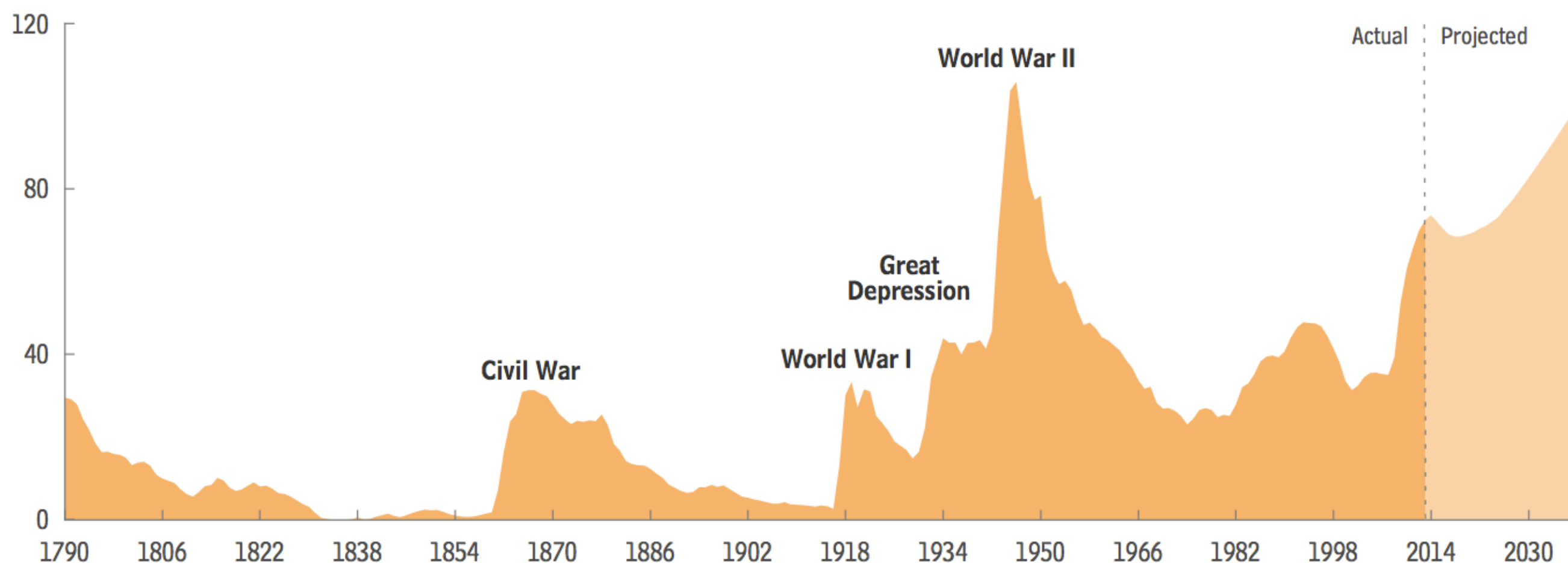
- What does the government tax?
- **Income**, since the 16th Amendment in 1913 and the Social Security Act of 1935
- So government's means are basically a function of income, and we will look at spending, taxes, and debt as a share of national income, **GDP**
- Seen in this light, government debt *doesn't look large*

U.S. history: Debt, warfare, and depression

Figure 1-1.

Federal Debt Held by the Public Under CBO's Extended Baseline

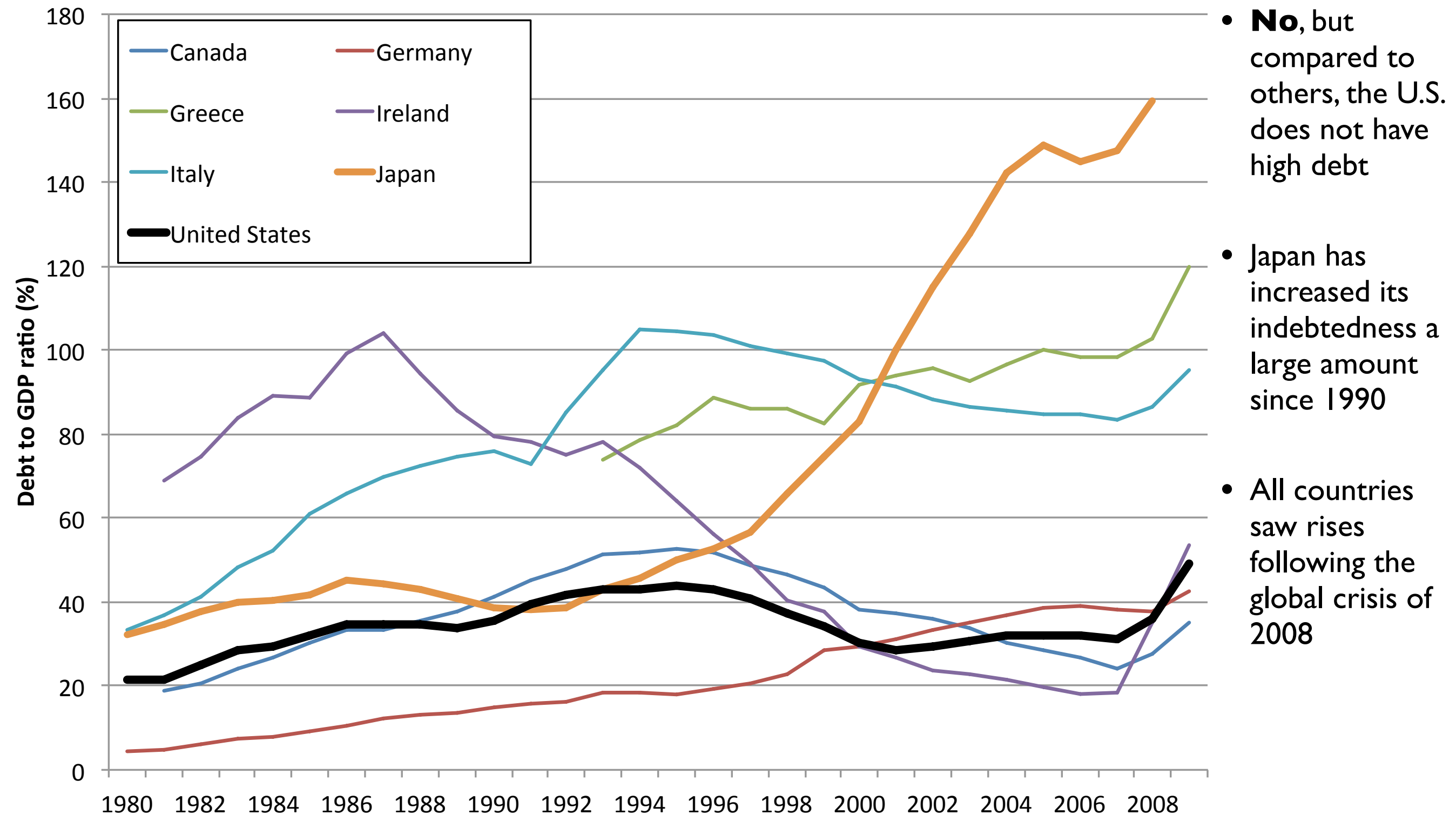
(Percentage of gross domestic product)



Source: Congressional Budget Office. For details about the sources of data used for past debt held by the public, see Congressional Budget Office, *Historical Data on Federal Debt Held by the Public* (July 2010), www.cbo.gov/publication/21728.

Notes: The extended baseline generally adheres closely to current law, following CBO's 10-year baseline budget projections through 2023 and then extending the baseline concept for the rest of the long-term projection period. The long-term projections of debt do not reflect the economic effects of the policies underlying the extended baseline. (For an analysis of those effects and their impact on debt, see Chapter 6.)

Does this mean there's no limit to spending?



- **No**, but compared to others, the U.S. does not have high debt
- Japan has increased its indebtedness a large amount since 1990
- All countries saw rises following the global crisis of 2008

The Financial Crisis worsened the fiscal outlook

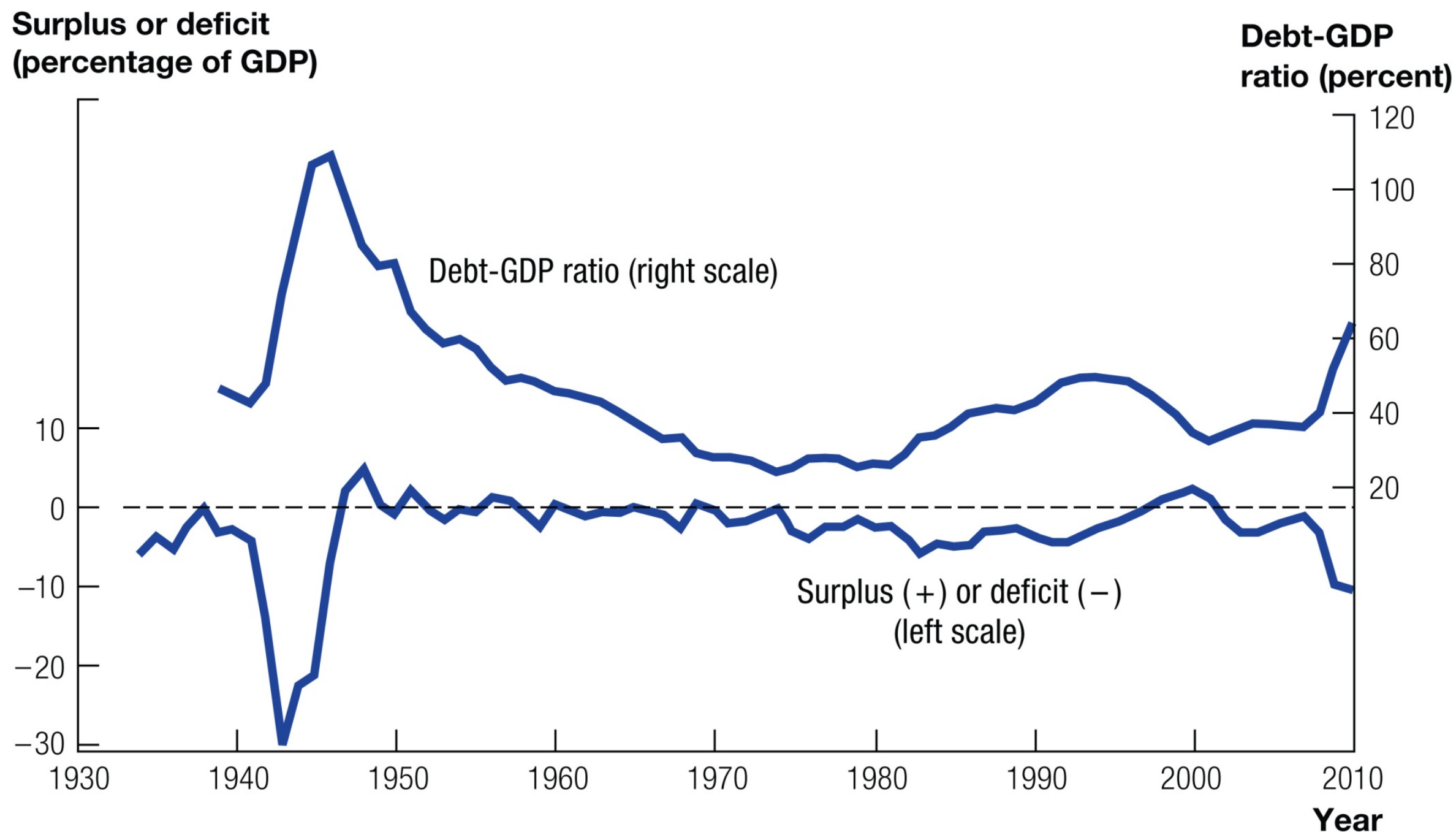


FIGURE 17.2 Federal Debt and Deficits in the United States, 1934–2010

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The recession and policy responses increased spending and reduced taxes

Percentage of GDP

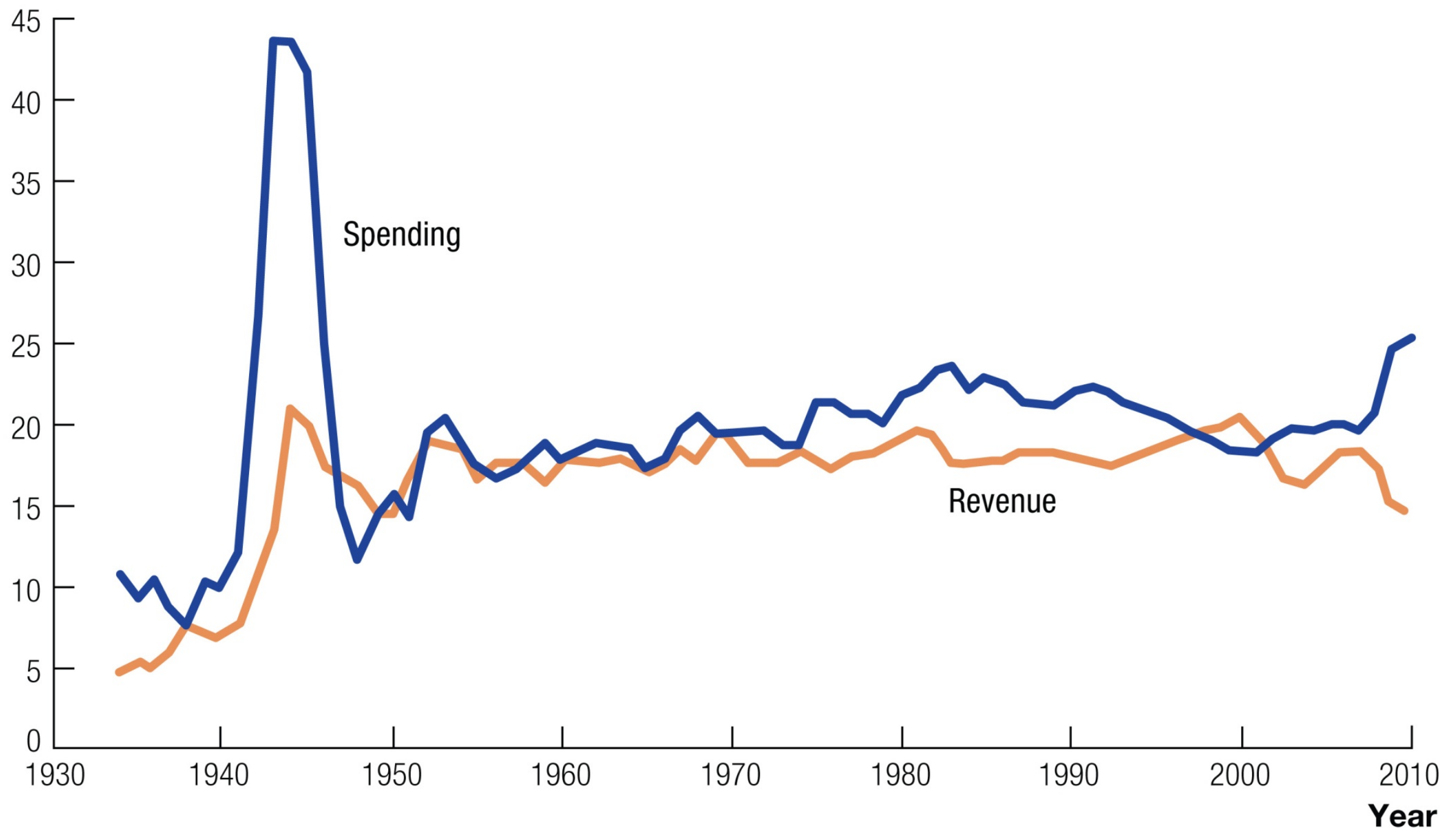


FIGURE 17.1 U.S. Federal Government Revenue and Spending, 1934–2010

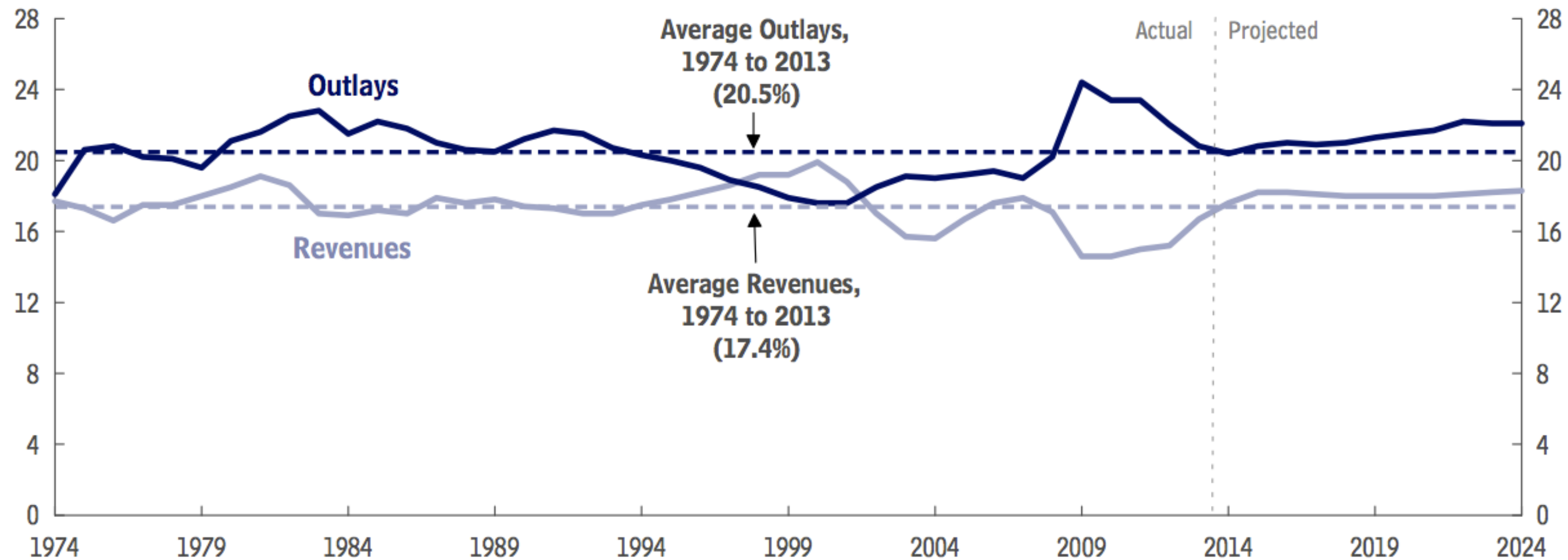
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April 2014 update from the Congressional Budget Office: Back from the brink, but no relief in sight

Figure 2.

Total Revenues and Outlays

(Percentage of gross domestic product)



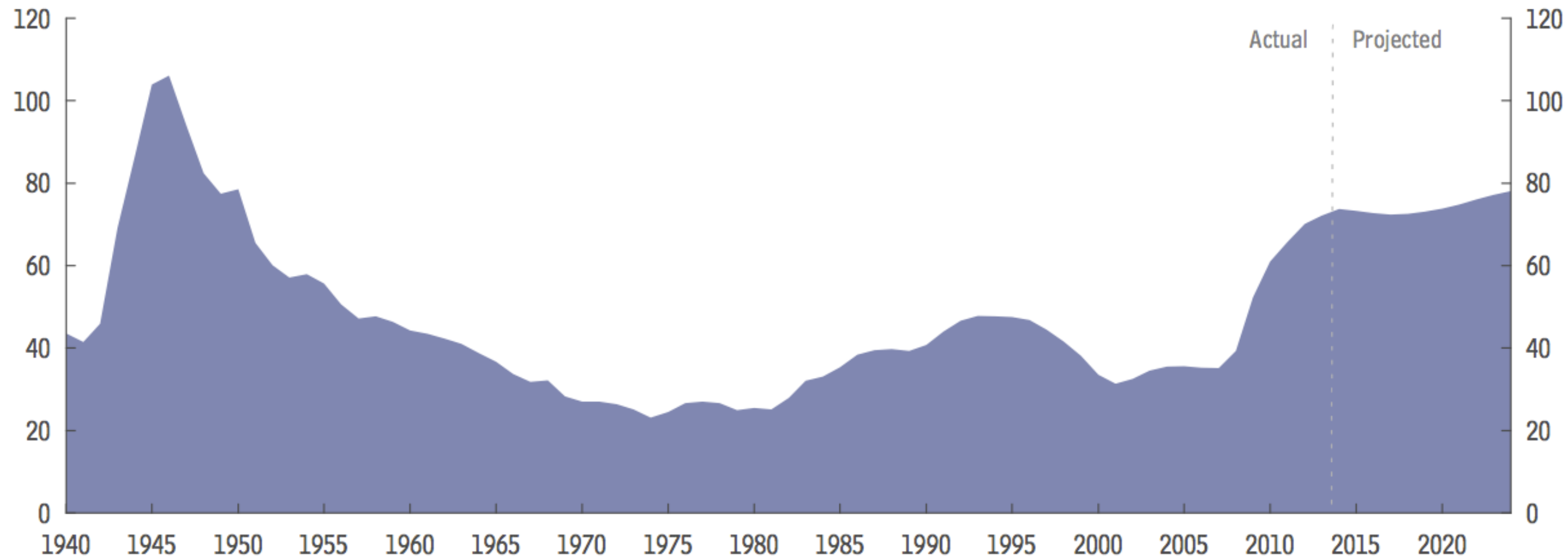
Source: Congressional Budget Office.

April 2014 update from the Congressional Budget Office: Back from the brink, but no relief in sight

Figure 3.

Federal Debt Held by the Public

(Percentage of gross domestic product)



Source: Congressional Budget Office.

How much of the deterioration in finances was due to the recession? How much was due to the policy response?

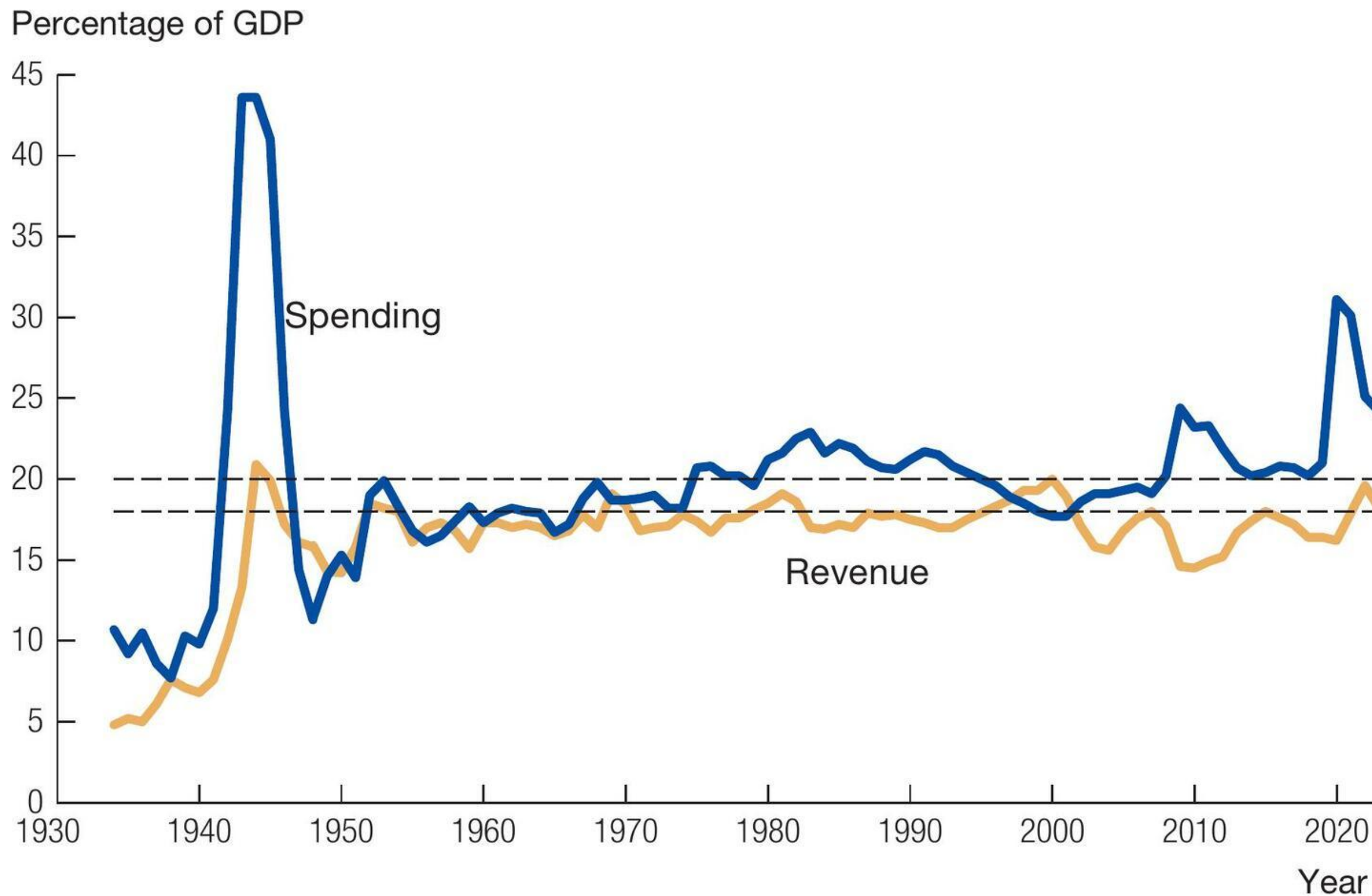
- When the economy enters a recession, four things can happen:
 1. Monetary policy responds. *This doesn't affect government finances*
 2. Government transfers *automatically rise* because some are “means-tested,” or tied to income, like food stamps
 3. Government taxes *automatically fall* because spending and income both fall, *and lower incomes are taxed at lower rates*
 4. Congress & the President can **decide** to cut taxes or raise transfers; or it can raise government purchases “G” (in $Y = C + I + \underline{\mathbf{G}} + NX$)
- How much of each of these happened during the Great Recession? Let's forget about #1, lump #2 and #3 together (“cyclical”), and compare to #4 (“policy”)

How much of the deterioration in finances was due to the recession? How much was due to the policy response?

	2007	2008	2009	2010	2011	2012
Total deficit as a percent of GDP	1.2%	3.2%	10%	8.9%	10.9%	7%
Cyclical share: due to expansion (+) or recession (–)	–0.7%	–0.1%	2.5%	3.3%	3.4%	3.3%
Structural share due to policy = the residual	1.8%	3.3%	7.5%	5.6%	7.6%	3.6%

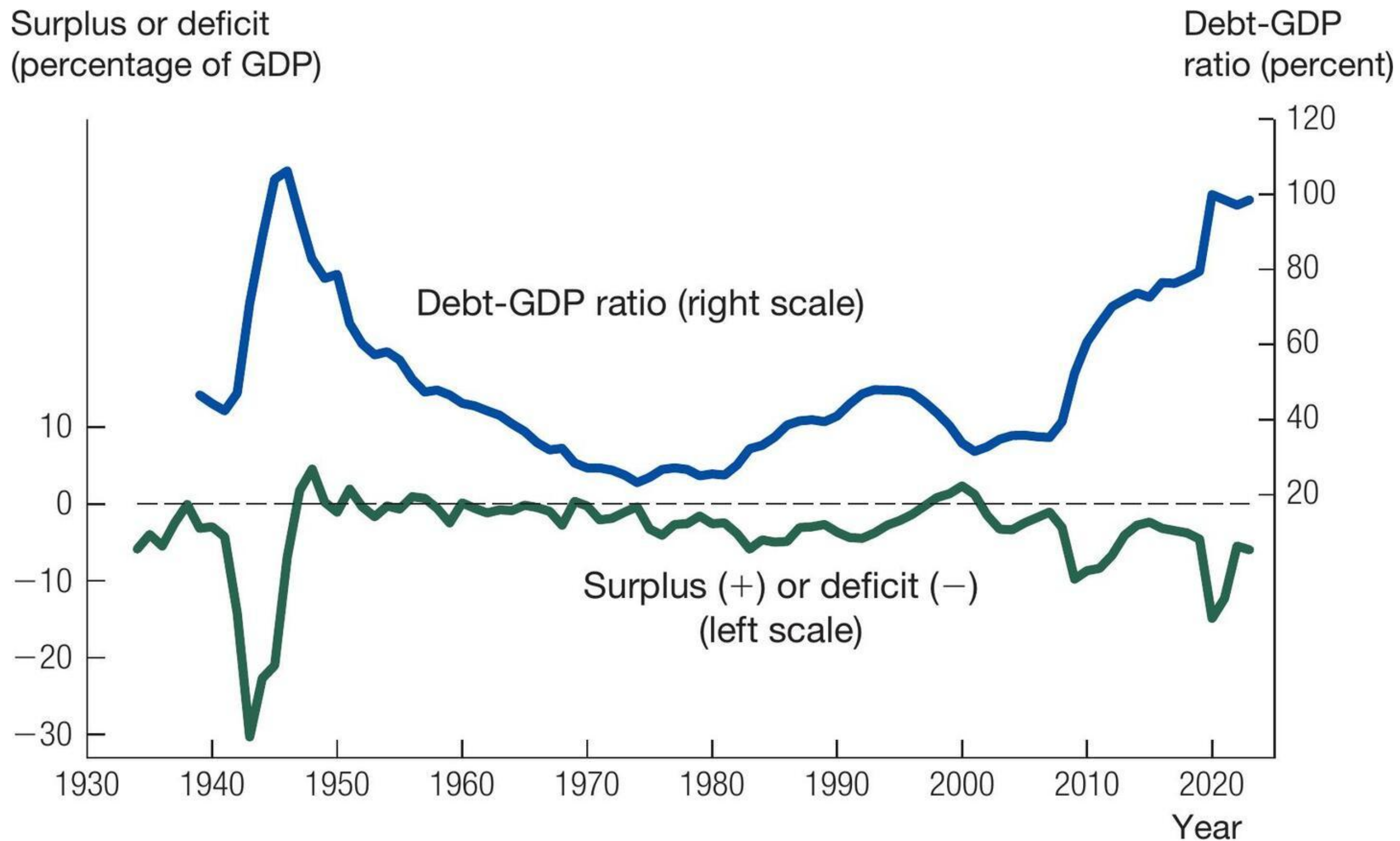
Source: President's FY2012 Budget, Table 3-4

Since the Great Recession, not a picture of fiscal balance



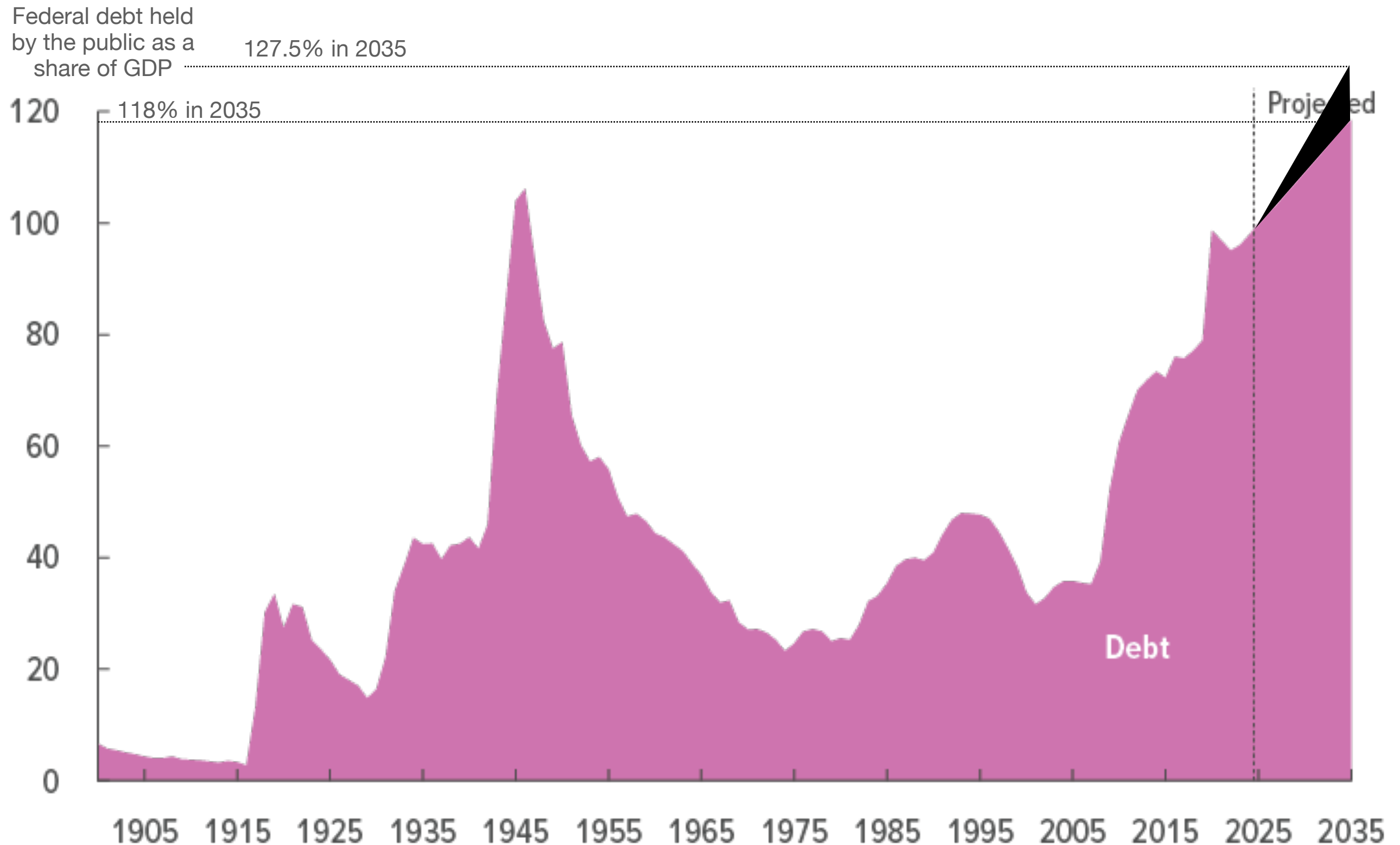
Source: *Economic Report of the President*, 2023, Table B-46.

Since the Great Recession, not a picture of fiscal balance



Source: *Economic Report of the President*, 2023, Table B-46.

CBO estimates that the July 2025 Reconciliation (“Big Beautiful Bill”) will further raise debt-GDP by 9.5 ppts at the end of 2034



What imposes limits to spending and debt?

- Although it can save or borrow in any one period, the government must adhere to a *budget constraint* **over time**, just like the rest of us
 - The textbook explores the budget constraint using **present value**
- Lenders know that high debt may prompt governments to *print money* and collect seignorage, which would cause inflation and erode the real value of their debt, so lenders may stop lending to high-debt countries
- Continued borrowing may place high very high tax burdens on future generations, our children and grandchildren
- ✓ Important: Borrowing may *crowd out investment*, lowering GDP

What is the government's **budget constraint**?

- In any one time period, the government can spend more than it takes in tax revenue by borrowing money from lenders, but it must eventually pay it back

$$G_t + Tr_t + iB_t = T_t + \Delta B_t + \Delta M_t$$

Spending
on goods &
services

Transfers
like Social
Security

Interest (i) on
outstanding
debt (B_t)

Taxes

Borrowing
new debt

Seignorage:
just printing
money

See the textbook for details.

What's important are the **present values** of spending and taxes today & tomorrow

Another limit: Excessive borrowing may crowd out investment and hurt GDP

- The government borrows money by offering Treasury bills, notes, and bonds to us, the **public**
- By buying a bond, we are “saving” in that we aren’t spending that money
- But we can’t simultaneously save that money and **invest** it in capital
- It’s an either/or kind of thing
 - Either your saving is investment that increases the capital stock
 - Or your saving finances the government’s deficit
 - (Unless you decide to save more and spend less)
- And there’s also the issue of **foreign saving** too ...

How do we see this? Start with income accounting:

$$Y = C + I + G + EX - IM$$

Now rearrange, putting investment, I , on one side:

$$I = Y - C - G - EX + IM$$

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Taxes, T , are subtractions from your income. So let's add and subtract T from the right hand side, and flip EX and IM:

$$I = Y - C - T + T - G + IM - EX$$

How do we see this? Start with income accounting:

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Now rearrange, putting investment, I , on one side:

$$I = Y - C - G - EX + IM$$

Taxes, T , are subtractions from your income. So let's add and subtract T from the right hand side, and flip EX and IM:

$$I = \underbrace{Y - C - T}_{\text{Income minus consumption minus taxes is what you have left for private saving}} + \underbrace{T - G}_{\text{Taxes minus spending equals public saving}} + \underbrace{IM - EX}_{\text{Imports minus exports is foreign saving (think: imported investment goods)}}$$

Income minus consumption
minus taxes is what you
have left for **private
saving**

Taxes minus
spending equals
public saving

Imports minus exports
is **foreign saving**
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Income minus consumption minus taxes is what you have left for **private saving**

Taxes minus spending equals **public saving**

Imports minus exports is **foreign saving** (think: imported investment goods)

Implications:

- If government spending G rises, what happens to investment I ? It depends...
- It depends on what happens to private and foreign saving
- Suppose private and foreign saving remain constant
- Then an increase in G without an increase in T will lower public saving, and that reduces investment
- Coming up:
What happens when foreign saving, a.k.a. the trade deficit $IM - EX$ rises?

Investment had closely tracked government saving ...

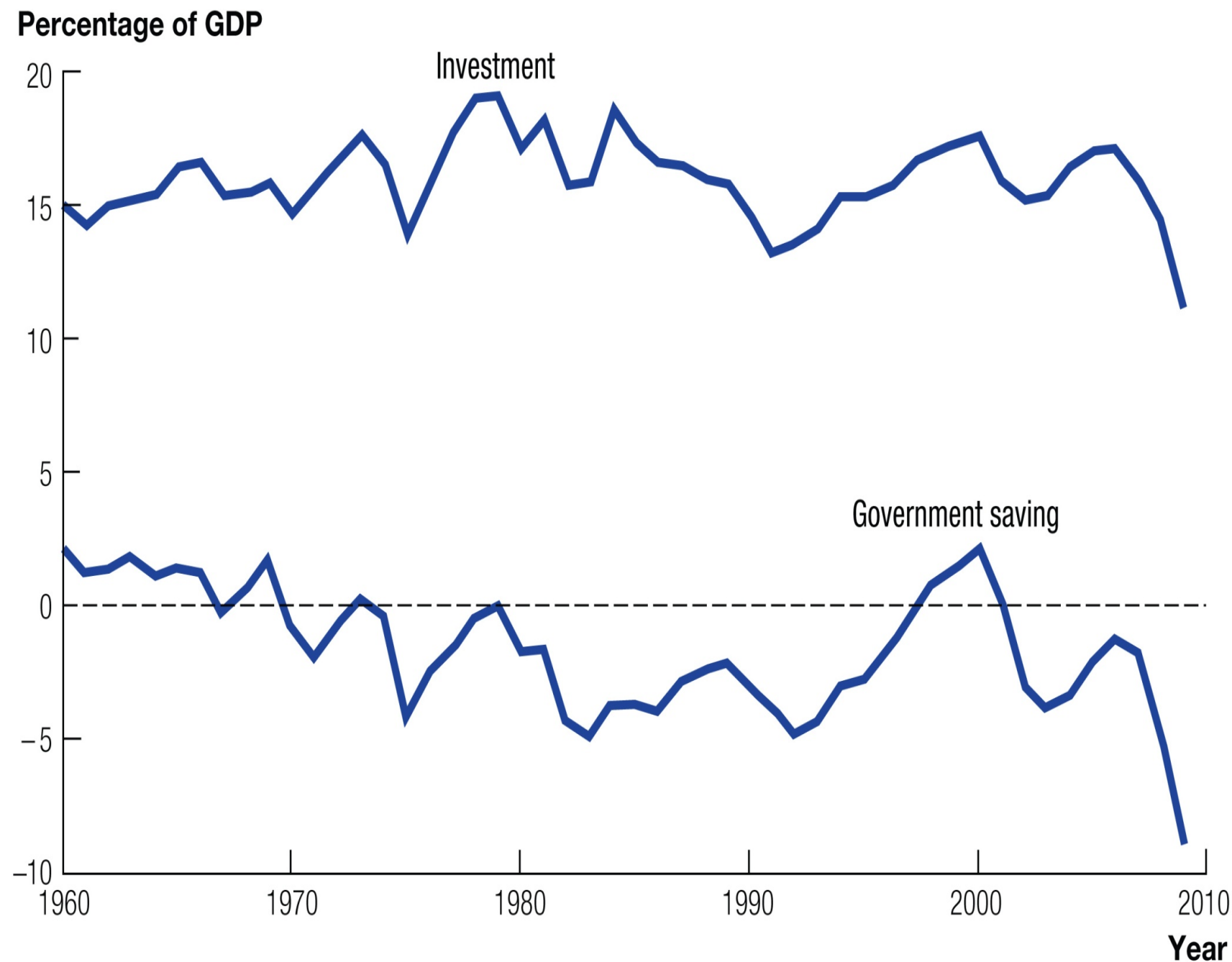
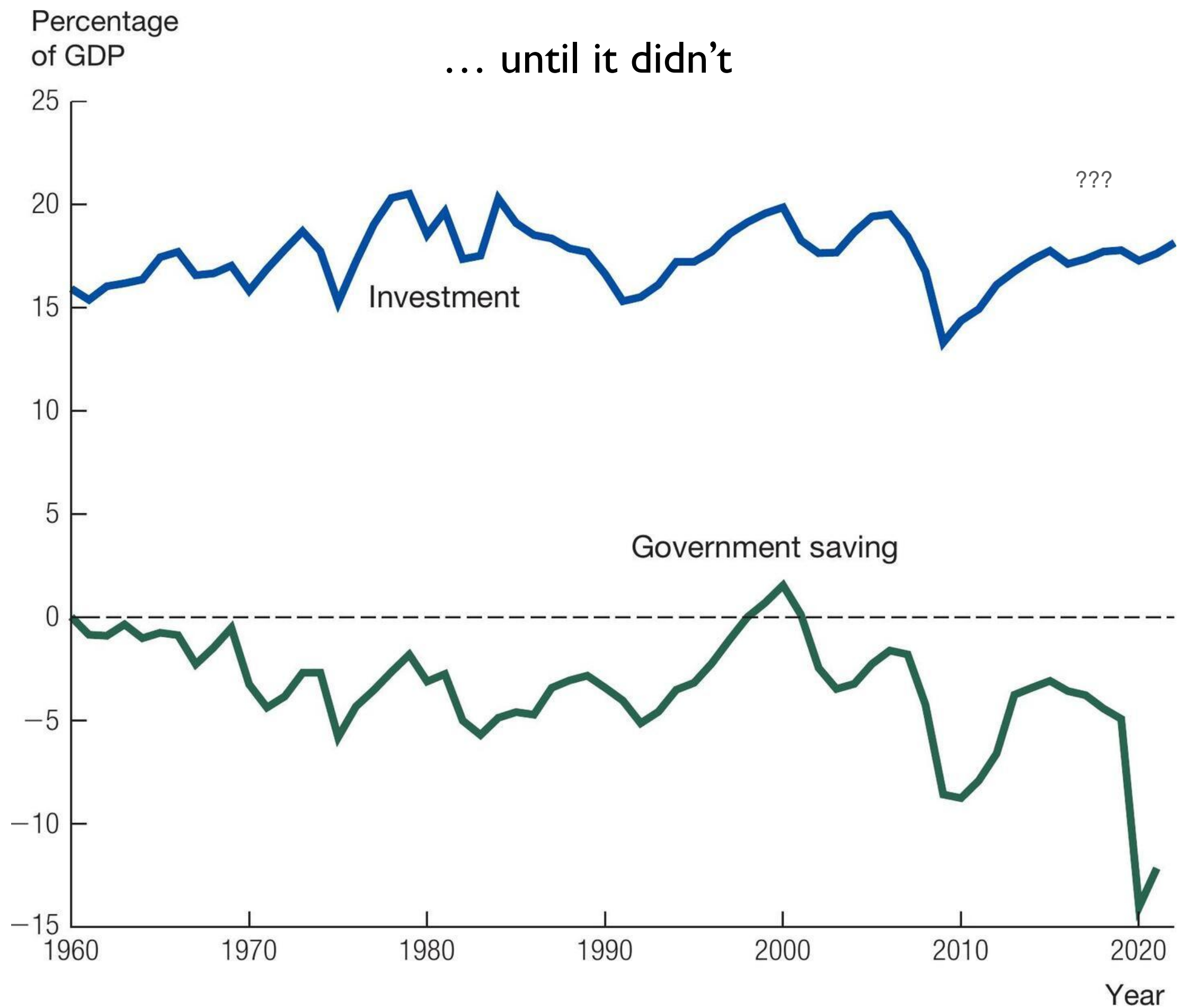


FIGURE 17.5 U.S. Investment and the Budget Deficit, 1960–2009

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Source: The FRED database.

Crowding out investment is bad, but doesn't the government also invest?

- Yes, some government activities qualify as investment in productive resources that will be useful for many years. Examples include:
 - **Roadways:** In 1956, Eisenhower signed the National Interstate and Defense Highways Act, which envisioned \$41 billion (about \$500 billion in 2026 dollars) for over 40,000 miles of new interstate
 - **Defense:** Although there is the potential for misuse, training and equipment for national defense pays clear dividends later
 - **Education:** We have seen how investment in human capital is important, and states and localities spend about \$1 trillion annually on K-12 education ([NCES Fast Facts](#))
 - **Health?** Human capital can depreciate either through skill obsolescence or through sickness and death! But does government health spending intervene early enough to matter?
- But a very large role for government is **social insurance** through transfer payments with less clear long-term payoffs

Composition of federal spending and taxes in 2005

	Billions of dollars	Dollars per person	Percent of GDP
Total Expenditures	\$2,433	\$8,330	19.8%
Health (+Medicare)	541	1,850	4.4
Social Security	516	1,760	4.2
National Defense	492	1,670	4
Income Security	344	1,170	2.8
Net Interest	184	620	1.5
Other	369	1,260	3
Total Revenues	\$2,126	\$7,260	17.3%
Income taxes	909	3,130	7.4
Payroll taxes	787	2,680	6.4
Corporate taxes	270	940	2.2
Other	147	520	1.2
Deficit (—)	—\$320	—\$1,070	—2.6%

Composition of federal spending and taxes in 2008

	Percentage of GDP	Dollars per person
Total expenditures	20.7	9,800
Health (including Medicare)	4.6	2,200
Social Security	4.3	2,030
National defense	4.3	2,020
Income security	3.0	1,420
Net interest	1.8	830
Other	2.7	1,290
Total revenues	17.5	8,290
Individual income taxes	7.9	3,760
Social insurance and retirement receipts	6.2	2,960
Corporate income taxes	2.1	1,000
Other	1.2	570
Budget deficit	3.2	1,510

TABLE 17.1 The U.S. Federal Government Budget, 2008

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TABLE 18.1

The U.S. Federal Government Budget, 2023

	Percentage of GDP	Dollars per person
Total expenditures	23.9	19,100
Health (including Medicare)	6.4	5,100
Social Security	5.1	4,000
National defense	3.1	2,400
Income security	3.0	2,400
Net interest	2.5	2,000
Other	3.9	3,100
Total revenues	18.0	14,400
Individual income taxes	8.7	7,000
Social insurance and retirement receipts	6.3	5,000
Corporate income taxes	2.0	1,600
Other	0.9	800
Budget deficit	-5.9	-4,700

Source: *Economic Report of the President, 2023*, Table B-47.

What are the major entitlement programs, and how do they work?

- **Social Security**

- Pays monthly pensions to retirees over the age of 62 who contributed their payroll taxes to the system back when working
- Receives your (and my) payroll taxes today in return for promises

- **Medicare**

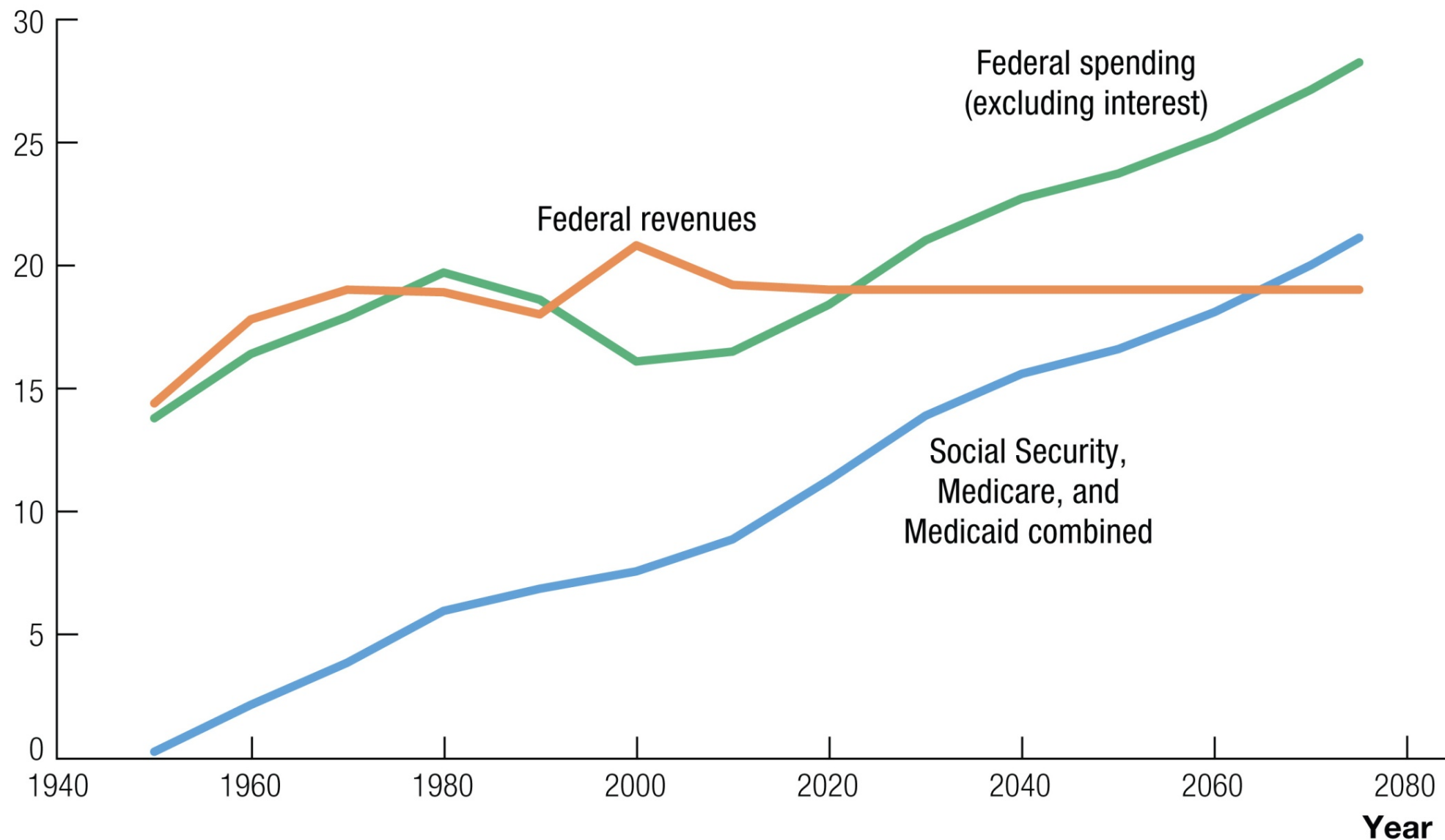
- Pays hospitals and subsidizes supplemental medical insurance and prescription drugs for Americans over 65
- Receives payroll taxes and some general revenues (income taxes)

- **Medicaid**

- Covers medical care for the poor and medically needy, including elderly nursing home (long-term care) patients
- Funded from general revenues

What is the outlook for all this spending on the elderly?

Percentage of GDP



- Not too good!
- While federal revenue, which is mostly the income tax, will remain fairly flat as a share of GDP,
- Federal noninterest or “primary” spending will grow rapidly, from about 15% of GDP today to near 30% by the year 2080
- Growth in primary spending is solely due to growth in entitlements

FIGURE 17.6 U.S. Federal Spending and Revenues, 1950–2075

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Contrary to popular belief, Medicare and Medicaid pose bigger problems than Social Security

Percentage of GDP

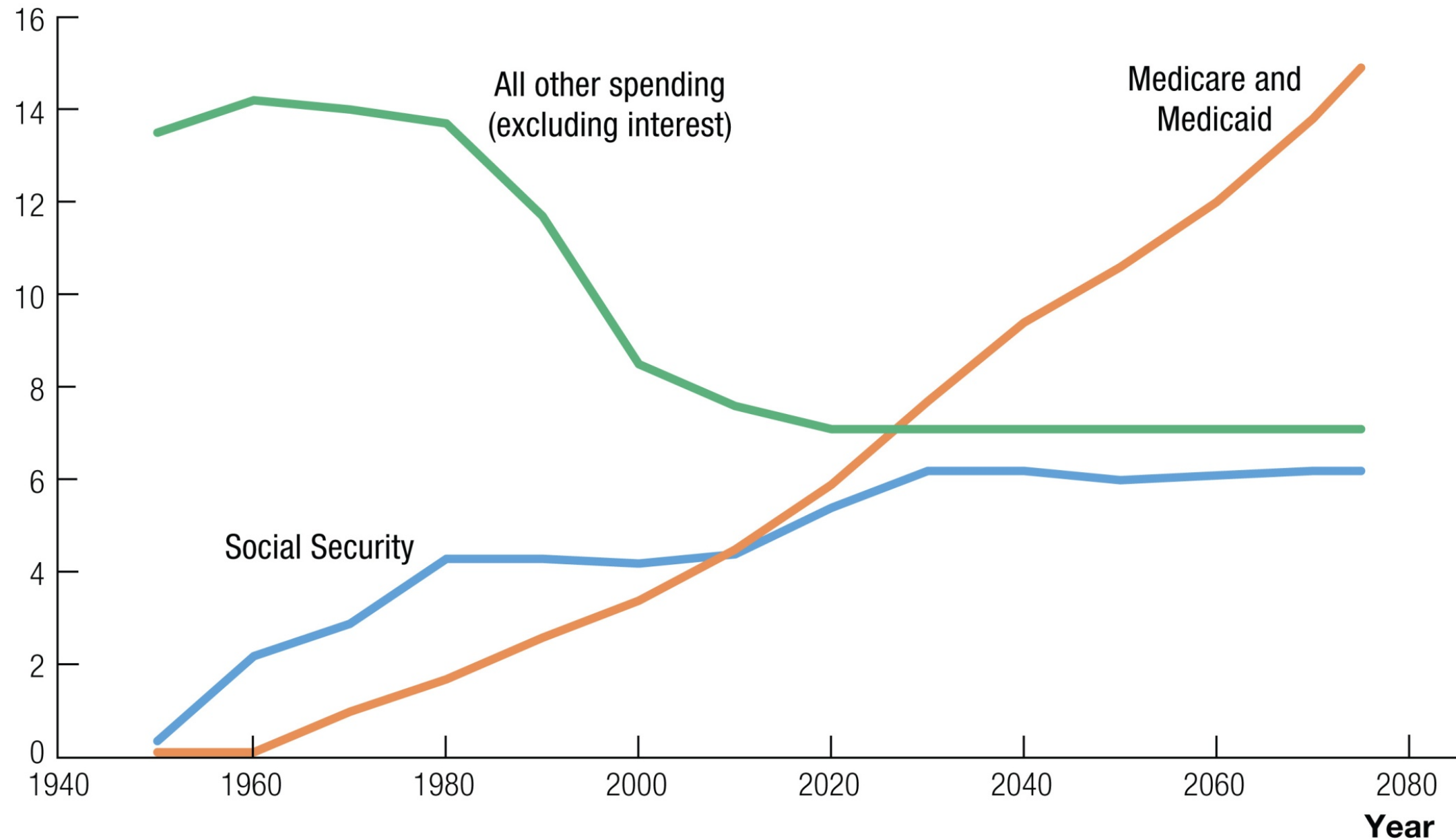
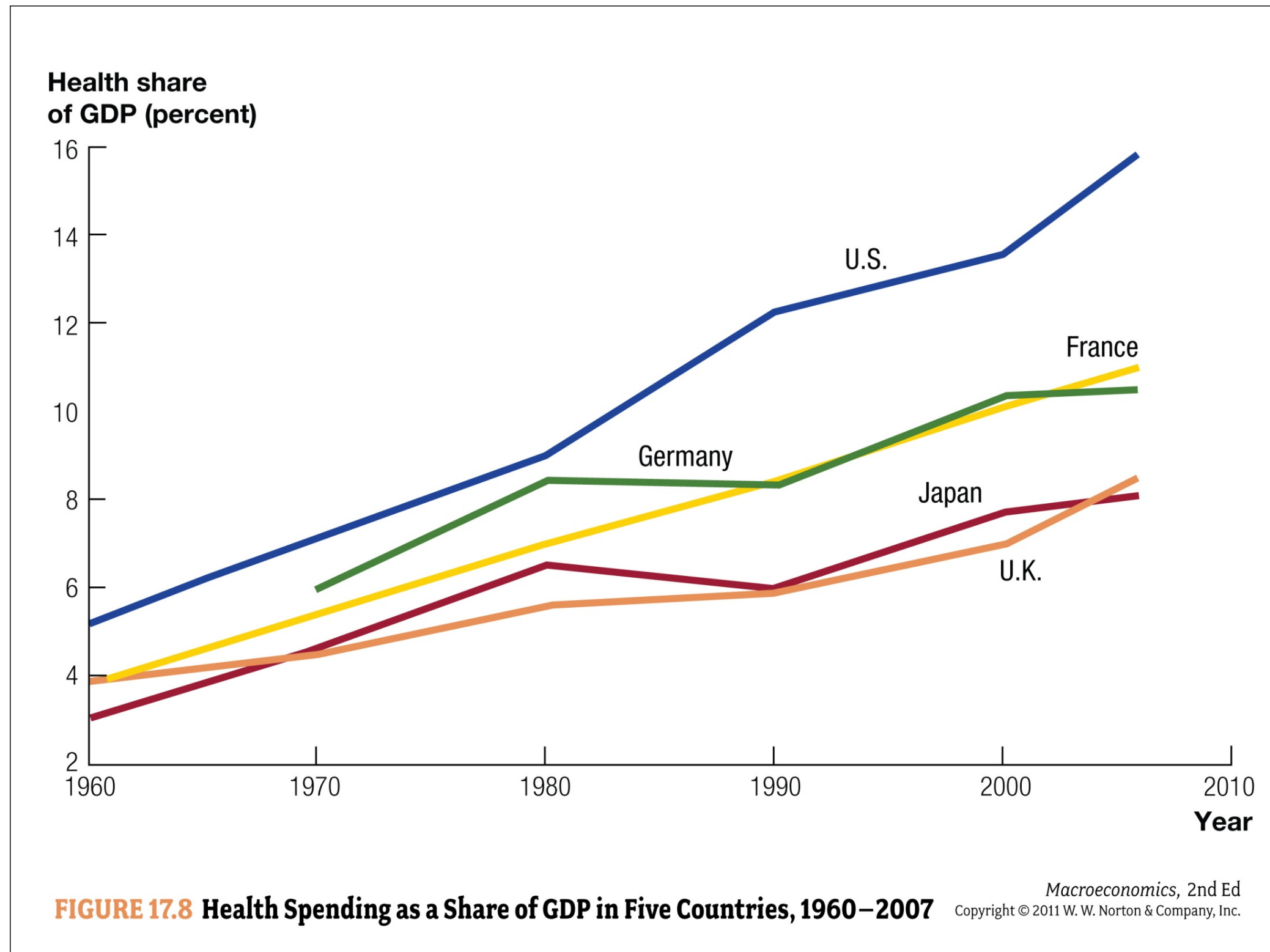


FIGURE 17.7 Components of Federal Spending, 1950–2075

Why is health spending rising so rapidly?

- Is there **waste** and **fraud** in Medicare and in the health system?
Yes, but probably not enough to drive such large, sustained increases
- Are people using too much medical care, since it's subsidized through health insurance?
Probably, but try telling that to your grandfather!
- Are doctors and hospitals ordering too many unnecessary procedures?
Probably, but try telling that to your grandfather!
- Is the U.S. subsidizing the development of prescription drugs?
*Probably, but **all** countries around the world find their health spending is rising as a share of their GDP — nobody's getting a free lunch*
- What's left? Maybe people **want** to spend more and more of their incomes on health, so they can live longer and healthier lives

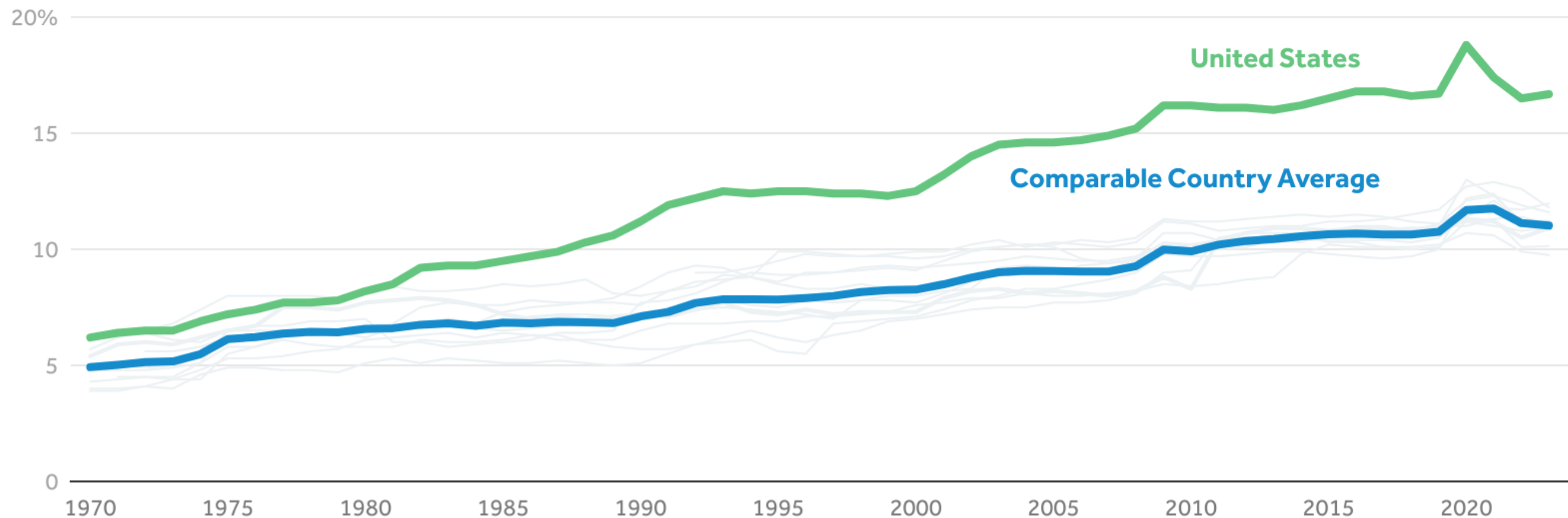
The share of GDP spent on health looks like it's rising everywhere ...



And it is, but looks can be deceiving

[Peterson-KFF Health System Tracker](#)

Health expenditures as percent of GDP, 1970-2023



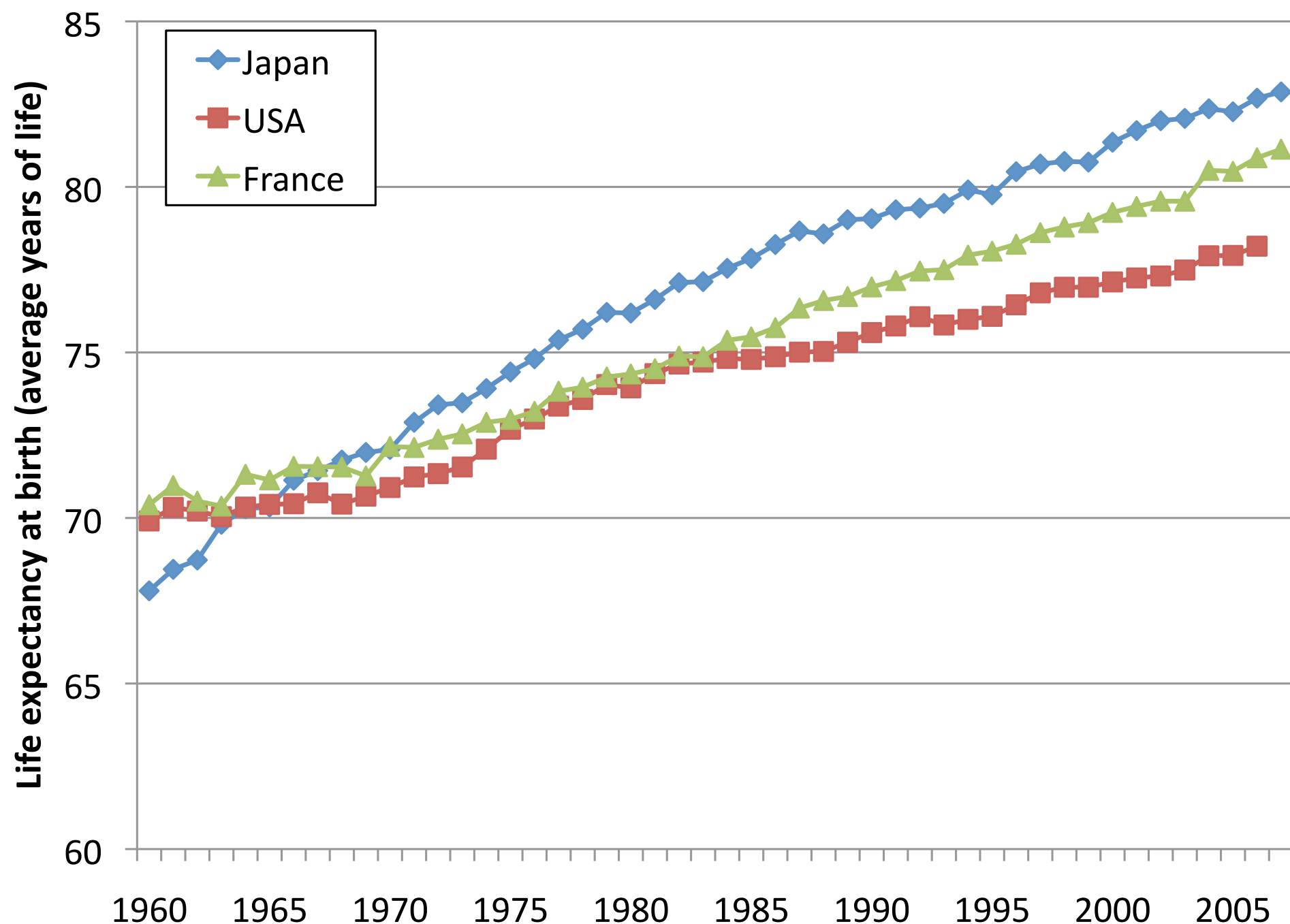
Notes: Data for 2023 from Australia, Belgium, Japan, the Netherlands, Switzerland, and United States are estimated. Data for 2023 from Austria, Canada, France, Germany, Sweden and the United Kingdom are provisional. Data for 2022 for Australia, Canada and Japan are provisional. Data for Australia is unavailable in 1970. Data for France from before 1990 is not available. Data from Germany prior to 1992 refers to West Germany. Data for Germany is not available for 1991. Data for the Netherlands is unavailable in 1970 and 1971. Originally published in: "How does health spending in the U.S. compare to other countries?"

Source KFF analysis of OECD data

Peterson-KFF

Health System Tracker

Average life spans — *life expectancies* — have been increasing rapidly here and abroad



- Like GDP per capita, length of life per capita has also risen
- More life is probably better than more consumption in any year of life
- (But why does it seem the U.S. is lagging behind?)

2010 Affordable Care Act a.k.a. “Obamacare”

- Aimed at expanding market-based health insurance coverage to Americans on the “individual market” — under age 65 without coverage provided through an employer
- For current college students:
 - You can stay on your parents’ plan until you turn 26yo
 - You can buy a [CoveredCA.com](https://www.coveredca.com) plan
 - CA and other states charge income tax penalties if you have no insurance. The federal government no longer does as of 2019
- Obamacare also contained **cost control mechanisms**
 - Center for Medicare and Medicaid Services (CMS) and [CMS Innovation Center](#) and [Accountable Care Organizations](#)
 - ACOs are groupings of doctors, hospitals, etc. that identify high-quality care *and share in any cost savings associated with cutting low-quality care*

Waste and fraud(?) in the news

"All the News That's Fit to Print"

The New York Times

Late Edition

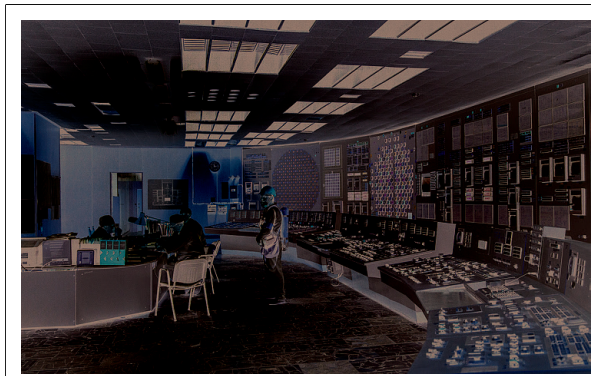
Today, partly sunny, seasonable, high 64. Tonight, turning cloudy, a bit of rain late, low 48. Tomorrow, cloudy, a little rain, breezy, cooler, high 54. Weather map, Page D8.

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\$2.50



Workers in the control room of the Unit 2 reactor, now closed, in Chernobyl, Ukraine, 28 years after the 1986 meltdown.

Putting a 32,000-Ton Cap on Chernobyl

By HENRY FOUNTAIN

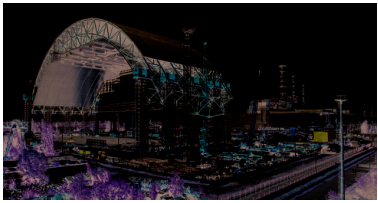
CHERNOBYL, Ukraine — Against the decaying skyline here, a one-of-a-kind engineering project is rising near the remains of the world's worst civilian nuclear disaster.

An army of workers, shielded from radiation by thick concrete slabs, is constructing a huge arch, sheathed in acres of gleaming stainless steel and vast enough to cover the Statue of Liberty. The structure is so otherworldly it looks as if it could have been dropped by aliens onto this Soviet-era industrial landscape.

If all goes as planned, by 2017 the 32,000-ton arch will be delicately pushed on Teflon pads to cover the ramsack shelter that was built to entomb the radioactive remains of the reactor that exploded and burned here in April 1986. When its ends are closed, it will be able to contain any radioactive dust should the aging shelter collapse.

By all but eliminating the risk of additional atmospheric contamination, the arch will remove the lingering threat of even a limited reprise of those nightmare days 28 years ago, when radioactive fallout poisoned the flatlands for miles around and turned villages into ghost towns, filled with the echoes of abandoned lives.

The arch will also allow the final stage of



An arch projected to cost \$1.5 billion is designed to all but eliminate the risk of further contamination at the site of the world's worst civilian nuclear disaster.

the Chernobyl cleanup to begin — an arduous task to remove the heavily contaminated reactor debris for permanent safe storage. That this job will fall from international hands to those of Ukraine presents new worries, especially as Russia threatens the nation's borders.

For now, though, the rising arch is a sign of progress.

"It's an amazing structure," said Nicolas Caille, project director for Novarka, the

consortium of French construction companies that is building it. "You can't compare it to anything else."

With nations debating the future of atomic power as one way to reduce greenhouse-gas emissions and fight climate change, the arch is also a stark reminder that nuclear energy, for all of its benefits, carries enormous risks. When things go wrong, huge challenges follow.

Continued on Page A10

One Therapist, \$4 Million In 2012 Medicare Billing

8 of Program's Top 10 Earners in Physical Therapy Practice in New York

By JULIE CRESWELL and ROBERT GEBELOFF

A few miles from the Coney Island boardwalk in Brooklyn stands an outpost of what, on paper, is a giant of American medicine.

Nothing about the place hints at the money that is said to flow there. But in 2012, according to federal data, \$4.1 million from Medicare coursed through the office in a modest white house on Ocean Avenue.

In all, the practice treated around 1,950 Medicare patients that year. On average, it was paid by Medicare for 94 separate procedures for each one. That works out to about 183,000 treatments a year, 500 a day, 21 an hour.

What makes these figures more remarkable, and raises eyebrows among medical experts, is that judging by Medicare billing records, one person did it all. His name is Wiel Bakry, and he is not some A-list cardiologist, oncologist or internist. He is a physical therapist.

But physical therapy, it turns out, is a big recipient of national Medicare dollars — and physical therapy in Brooklyn is among the biggest of all. Of the 10 physical therapists nationwide who were paid the most by Medicare in 2012, half listed Brooklyn addresses, according to an analysis of Medicare billing data by The New York Times. Two others listed addresses on Long Island, one in Queens, and one each in California and Texas.

Mr. Bakry, 42, did not dispute the \$4.1 million figure. But he said that the Medicare data, released to much fanfare this month, could be "deceiving." One person, he said, could not provide all of that care alone.

"I'm not Superman," Mr. Bakry said.

One thing is certain: Physical therapy has become a Medicare gold mine. Medicare paid physical therapists working in offices \$18.1 billion in 2012 alone, the 10th-highest field among 74 specialties, according to the Times analysis. In Brooklyn, physical therapy was second only to internal medicine.

Why Brooklyn? Federal authorities say the borough is a na-

A WHITE HOUSE SPLIT OVER RUSSIA

Issue Is Whether to Be Tougher Than E.U.

By PETER BAKER and C. J. CHIVERS

WASHINGTON — As President Obama and his national security team struggle to increase pressure on Russia over its intervention in Ukraine, they have become entangled in a tense debate over how much emphasis to put on unity with European allies more reluctant to take stronger economic actions against Moscow.

So far, Mr. Obama has opted to stick close to the Europeans to maintain an undivided front, even at the expense of more punishing sanctions and quicker responses to Kremlin provocations. But some inside and outside the administration argue that the United States should act unilaterally if necessary, on the assumption that the Europeans will ultimately follow.

The issue came to a head in recent days as American and European leaders tried to coordinate a new round of sanctions after the collapse of a Geneva agreement to de-escalate the crisis in Ukraine. Secretary of State John Kerry delivered a blistering public attack on Moscow on Thursday night for not living up to the agreement, but the plan to follow that up with sanctions on Friday fell apart while Washington waited for Europe, postponing action until Monday at the earliest.

The deliberations in the West came as pro-Russian forces in Ukraine on Sunday continued to defy international demands to stand down. An antigovernment militia paraded eight detained members of a European military observer mission before cameras, while protesters seized a regional government television station and declared they would use it to air Russian newscasts.

The display of the captive European observers underscored the challenge for Washington and

Gay Rights Push Shifts Its Focus South and West

By NICHOLAS CONFESSORE and JEREMY W. PETERS

The country's leading gay rights groups and donors, after a decade focused on legalizing same-sex marriage, are embarking on a major drive to win more basic civil rights and workplace protections in Southern and Western states where the rapid progress of the movement has largely eluded millions of gay men and lesbians.

The effort will shift tens of millions of dollars in the next few years to what advocates described as the final frontier for gay rights: states like Mississippi, Georgia, Arkansas and Texas, where Republicans dominate elected office and traditional cultural views on homosexuality still prevail.

The new strategy reflects the growing worry within the movement that recent legal and political successes have formed two quickly diverging worlds for lesbian, gay, bisexual or transgender Americans: one centered on the coasts and major cities, and

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Amid Uproar, Clippers Silently Display Solidarity

By JOHN BRANCH

OAKLAND, Calif. — The Los Angeles Clippers, reacting to recordings of racist remarks attributed to Donald Sterling, the team's owner, took the court for their playoff game Sunday with a statement, one of both fashion and politics.

In a silent sign of solidarity, players shed their warm-up jackets before the game and placed them in a pile at midcourt, revealing red shirts worn inside out to obscure the team's name. And while they wore the Clippers' blue jerseys during the game, each player also wore black socks with black wristbands.

With her husband barred from attending the game while the N.B.A. investigates the remarks, Mr. Sterling's wife of more than 50 years, Rochelle, sat courtside, where Republicans dominate elected office and traditional cultural views on homosexuality still prevail.

The recordings of the racist remarks bounced around the globe Sunday as the Clippers' bench, the league's leadership, overshadowing its weekend state of playoff games and even receiving the attention of President Obama in Malaysia.

But the epicenter was at Oracle



Los Angeles Clippers players during Sunday's game. Players wore their warm-up shirts inside out, obscuring the team name.

Arena, where the Clippers played the Golden State Warriors in Game 4 of their first-round playoff series. The Warriors jumped to a huge first-quarter lead, on their way to a 118-87 win that tied the best-of-seven series at 2-2.

The comments attributed to Mr. Sterling have ignited a firestorm in the N.B.A., in which roughly three-quarters of the players are black and nearly every owner is white. That it should involve a team based in Los An-

geles, a diverse city with its own history of racial problems but one that has long revered black athletes, only added long threads of complex context.

Audio of the comments, first released by the website TMZ, seems to catch Mr. Sterling arguing with a female friend, telling her for posting pictures of herself online with black men, including the basketball great and

Continued on Page D3

In Venezuela, Protesters Point To Their Scars

By WILLIAM NEUMAN

VALENCIA, Venezuela — Clippers' Martinez was shot at such close range by a soldier at a protest that his surgeon said he had to remove pieces of the plastic shogun shell buried in his leg, along with the shards of keys Mr. Martinez had in his pocket, shattered by the blast.

Jordana Gregory was detained with 10 others who said that over three days they were kicked, pistol-whipped, doused with pepper spray and battered with helmets and shotgun bullets.

"They wanted to make people afraid so we wouldn't stay in the streets," said Mr. Gregory, 19. "But what happened was more protests and more deaths."

Venezuela has been shaken by more than two months of often violent protests that President Nicolas Maduro says are designed to overthrow him. He has held the opposition responsible for violence that the government says has claimed more than 40 lives, including those of protesters, bystanders and six National

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NATIONAL A12-14

A Mayor's Busy Life Online

By regularly posting photographs on Instagram, Mayor Eric Garcetti of Los Angeles is giving an inside view of his routine and showing how politicians can use social media in new ways. PAGE A12

Constitution and Cellphones

The Supreme Court this week will consider whether the police need warrants to search the cellphones of people they arrest. PAGE A12

A Gun the Lobby Doesn't Love

A California company's effort to sell the nation's first "smart gun" has met stiff resistance from the gun lobby. PAGE A14

INTERNATIONAL A4-11

Pope Declares 2 New Saints

Before crowds in Rome, Pope Francis, right, with Benedict XVI, performed a canonization for his predecessors, John XXIII and John Paul II, whom he portrayed as "men of courage." PAGE A4



NEW YORK A15-19

Views of Immigrant Smuggler

A woman known as "Sister Ping" died in prison for her role in the 1993 Golden Venture disaster, but is remembered fondly in parts of Chinatown. PAGE A17

#NYPDLeadsLesson

An outreach campaign on Twitter stumbled, but it showed that the Police Department was experimenting. PAGE A17

BUSINESS DAY B1-7

Squeezing Cable TV's Profits

New challenges are chipping away at cable television's profitable bundles of channels, David Carr writes. PAGE B1

ARTS C1-7

Uncommon Visit to New York

Sculptures by the Renaissance artist Donatello, like "Lo Zuccone," will travel from the Duomo in Florence, Italy, to the Museum of Biblical Art in New York. PAGE C1

Hoping to Hear From Tony

The most recent revival of "The Glass Menagerie" could bring this Tennessee Williams play something it has never had — a Tony nomination. PAGE C1

SPORTSMONDAY D1-7

Rangers Regain Series Lead

A 4-2 win at Madison Square Garden put Rangers ahead of the Philadelphia Flyers, three games to two. PAGE D3

A Twice-in-a-Lifetime Chance

Art Sherman, who was an exercise rider for the 1955 Kentucky Derby winner, trains this year's favorite. PAGE D1

EDITORIAL, OP-ED A20-21

Paul Krugman

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